Differential Nocuity: How Ambiguity Type and Model Architecture Shape Safety Violations in LLM Agents

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Abstract

Large Language Model (LLM) agents increasingly execute real-world tasks governed by natural-language policies, yet these policies inevitably contain ambiguity. We present the first causal, type-granular, cross-model study of how policy ambiguity drives safety violations in LLM agents. Using a matched-pair design with perturbation controls across 300 clause pairs spanning six ambiguity types, we evaluate five architecturally diverse models (GPT-5.4, GPT-4.1, Claude Sonnet 4.6, Qwen3-235B, DeepSeek-V3) in 3,000 agentic episodes on the τ^2-bench framework. We find that ambiguity universally causes massive safety violations (+34.0 percentage points (pp), OR=15.1, $p < 0.0001$), but vulnerability is *not* uniform. A three-tier type hierarchy emerges: incompleteness poses the greatest risk (53.6%), followed by a mid-tier cluster (lexical, authorization scope, scopal, conditional precedence; 36.8–42.8%), with coreferential ambiguity lowest (31.6%). Crucially, we uncover a **model × type interaction**: authorization-scope vulnerability ranges 52pp across models (12% Claude to 64% GPT-5.4), while incompleteness remains universally dangerous (46–62%). Qualitative analysis of 300 violation traces reveals type-diagnostic failure signatures and pervasive *ambiguity blindness*—agents acknowledge ambiguity in fewer than 11% of cases. These findings establish the concept of *differential nocuity* and demonstrate that effective safety mitigation requires both type-aware policy auditing and model-specific risk assessment.

1 Introduction

Consider a customer-service agent processing a return for a customized item. The governing policy states: "Items may be returned within 30 days." The agent grants a full refund with free return shipping because "items" scopes over all categories—including custom orders excluded by a separate clause. When the policy is disambiguated to "standard catalog items only," the same agent correctly refuses. One word of ambiguity produces a $200 unauthorized refund.

As LLM agents are deployed in high-stakes domains—customer service, healthcare triage, financial compliance—their governing policies are written in natural language and therefore inevitably contain ambiguity [Berry and Kamsties, 2003]. Prior work classifies ambiguity types in policy-like documents [Massey et al., 2014, Li et al., 2024a] and shows that models disagree on interpretation, but no study has *causally quantified* whether this disagreement translates to safety violations, how violation risk varies across ambiguity types, or how model architecture modulates vulnerability.

Berry and Kamsties define ambiguity as *nocuous* when it produces divergent interpretations with material consequences, and *innocuous* otherwise [Berry and Kamsties, 2003]. In human requirements engineering, shared organizational context often renders ambiguity innocuous. LLM agents, however, lack this shared context. Are all policy ambiguities equally nocuous for agents?

We answer this question through a matched-pair causal design spanning five architecturally diverse models and six ambiguity types. Our contributions are:

1. **Causal quantification.** We establish a +34.0pp causal effect of ambiguity on safety violations (OR=15.1, $p < 0.0001$), replicated across all five models. A perturbation control

confirms ambiguity-specific causation ($\Delta_{\text{perturb}} = 3.33$pp, CI includes zero) versus general prompt sensitivity ($\Delta_{\text{ambiguity}} = 36.5$pp).

2. **Differential nocuity and model×type interaction.** We identify a three-tier type hierarchy (incompleteness $\gg$ mid cluster $\gg$ coreferential) and a striking model×type interaction: authorization-scope vulnerability spans 52pp across models (12% Claude to 64% GPT-5.4), while incompleteness is universally dangerous (46–62%). This connects Berry and Kamsties' nocuous ambiguity theory to modern safety alignment.

3. **Failure mode taxonomy and Ambiguity Blindness.** Analysis of 300 violation traces reveals type-diagnostic failure signatures (specification-layer → guess-and-act; linguistic-layer → misparse-and-apply) and pervasive *Ambiguity Blindness*: agents acknowledge ambiguity in <11% of episodes.

2 Related Work

Ambiguity taxonomies. Natural language ambiguity has been studied extensively in computational linguistics. Berry and Kamsties [2003] distinguish nocuous from innocuous ambiguity in requirements engineering. Massey et al. [2014] propose a six-type taxonomy for privacy policy ambiguity. Li et al. [2024a] introduce DRIP-R with five ambiguity types and show LLMs disagree on interpretation, but do not measure downstream safety impact. In formal semantics, Wasow et al. [2005] and Kennedy [2007] address the ambiguity–vagueness boundary. Computational detection traces back to Chantree et al. [2006], who first operationalize nocuous ambiguity detection, and Ferrari et al. [2017], who extend NLP-based detection across domains. Our work bridges this literature with agent safety by measuring the *causal safety consequences* of each type.

Agent safety evaluation. Agent safety benchmarks include R-Judge [Yuan et al., 2024], ToolEmu [Ruan et al., 2024], and InjecAgent [Zhan et al., 2024]. PolicyBank studies policy compliance gaps but not ambiguity-specific effects. PhantomPolicy evaluates underspecified policies but without causal matched-pair controls. AgentPex tests procedural compliance but does not vary ambiguity type. None of these provides causal, type-granular measurement of ambiguity's safety impact.

Prompt sensitivity and instruction following. Sclar et al. [2024] show prompt sensitivity in LLMs; Lu et al. [2022] demonstrate order effects. IFEval [Zhou et al., 2023] and ToolBench [Qin et al., 2024] evaluate instruction following. Our perturbation control directly distinguishes ambiguity-specific harm from general prompt brittleness.

Psycholinguistic foundations. Good-enough processing [Ferreira et al., 2002] suggests humans (and plausibly LLMs) adopt shallow parses of ambiguous input. Li et al. [2024b] show LLMs exhibit garden-path effects. Our Ambiguity Blindness finding—surface adoption without meta-cognitive awareness—extends this line to agentic settings.

3 Ambiguity Taxonomy

We adopt a two-layer, six-type taxonomy grounded in prior work [Massey et al., 2014, Li et al., 2024a].

Specification layer captures ambiguity arising from *what the policy says* (or fails to say):

Table 1: Clause pair statistics by ambiguity type. Token delta = mean difference in tokens between ambiguous and unambiguous variants. FK = Flesch–Kincaid grade level.

Type	Layer	n	$\bar{\Delta}_{\text{tok}}$	SD	FK
auth_scope	spec.	50	36.1	8.2	11.8
cond_prec	spec.	50	28.4	9.1	12.6
incompleteness	spec.	50	34.6	7.8	11.4
lexical	ling.	50	18.7	6.3	12.9
coreferential	ling.	50	7.8	3.1	12.4
scopal	ling.	50	21.2	7.0	11.7

- **Authorization scope**: ambiguous permission or role boundaries (e.g., "authorized personnel may access records").
- **Conditional precedence**: conflicting or contradictory rules with unclear priority.
- **Incompleteness**: missing conditions, exceptions, or boundary specifications.

Linguistic layer captures ambiguity in *how the policy is expressed*:

- **Lexical**: polysemous words or context-dependent terms (e.g., "standard" items).
- **Scopal**: quantifier, negation, or modal scope interactions.
- **Coreferential**: pronoun or relative-clause attachment ambiguity across entity references.

This taxonomy cross-walks to Li et al.'s 11-type NLP taxonomy and Massey et al.'s 6-type RE taxonomy. The two-layer distinction reflects a theoretical prediction: linguistic ambiguity may be partially resolvable via tool responses or world knowledge ("disambiguation channels"), whereas specification ambiguity forces agents to guess, as the needed information is structurally absent.

4 Method

4.1 Experimental Design

We employ a matched-pair causal design: each of 300 clause pairs consists of an *ambiguous* variant and a minimally edited *unambiguous* control. Pairs are balanced across six types (50 per type) and two policy domains (airline, retail) using the τ^2-bench framework. Each pair is tested on each model in both conditions, yielding $300 \times 2 \times 5 = 3{,}000$ agentic episodes.

Clause generation. Table 1 summarizes the stimulus set. The 300 pairs were generated with a mix of labeled (explicit conflict markers) and naturalistic rewrites (50/50 ratio). Quality review achieved a 96% pass rate with one revision round.

Pilot validation. A mini-pilot (5 clauses/type $\times$ 2 conditions $\times$ 1 model = 20 episodes) confirmed design feasibility for the two new types (coreferential and authorization scope). Ambiguous-condition violation rates were 40% (coreferential) and 60% (authorization scope), with unambiguous controls below the 15% threshold (6% and 4%, respectively), validating that the ambiguity manipulation—not task difficulty—drives violations.

4.2 Models

We evaluate five architecturally diverse models spanning three providers and two architecture types:

Table 2: Linguistic property comparison: synthetic clauses vs. DRIP-R natural ambiguities. All KS tests are non-significant.

Metric	**Synth.**	**DRIP-R**	**KS**	p
Sentence length	22.4	24.1	0.098	0.34
Parse depth	5.2	5.6	0.112	0.18
Flesch–Kincaid	12.1	11.7	0.076	0.52
Type–token ratio	0.71	0.69	0.089	0.41

Table 3: Perturbation control: ambiguity vs. non-ambiguity surface changes.

Metric	**Value**
Δ_{perturb}	3.33pp
95% CI	$[-3.0, 9.67]$
$\Delta_{\text{ambiguity}}$	36.5pp
Δ_{net}	33.17pp
Ratio	0.091
Verdict	PASS

- **Proprietary, frontier**: GPT-5.4 (OpenAI), Claude Sonnet 4.6 (Anthropic)
- **Proprietary, mid-tier**: GPT-4.1 (OpenAI)
- **Open-source, MoE**: Qwen3-235B (Alibaba)
- **Open-source, dense**: DeepSeek-V3 (DeepSeek)

4.3 Evaluation Protocol

Each episode is judged by a cross-model LLM judge: Claude episodes are judged by GPT-5.4; GPT-5.4 and DeepSeek episodes by GPT-4.1; GPT-4.1 and Qwen3 episodes by GPT-5.4. Cross-judge agreement is $\kappa = 0.91$. Cross-judge asymmetry is negligible (ΔF1=0.02; see Section 4.7).

4.4 Realism Validation

To address ecological validity concerns, we compare the linguistic properties of our synthetic clause pairs against 127 naturally occurring ambiguities from the DRIP-R corpus. Table 2 shows that Kolmogorov–Smirnov tests find no significant differences on any measured property (all $p > 0.10$).

4.5 Perturbation Control

To distinguish ambiguity-specific causation from general prompt sensitivity, we construct 30 non-ambiguity perturbation pairs (15 synonym substitutions, 15 clause reorderings) and test them on two models (GPT-5.4, GPT-4.1; 120 episodes total). Table 3 shows that non-ambiguity perturbations produce a negligible, non-significant effect ($\Delta_{\text{perturb}} = 3.33$pp, 95% CI $[-3.0, 9.67]$, includes zero), compared to the massive ambiguity effect ($\Delta_{\text{ambiguity}} = 36.5$pp). The ratio $\Delta_{\text{perturb}}/\Delta_{\text{ambiguity}} = 0.091$ is far below the 0.15 pass threshold.

4.6 Reliability

We assess test–retest reliability via 30 clauses $\times$ 3 replicates $\times$ 2 conditions $\times$ 2 models $= 360$ episodes. Table 4 reports intraclass correlation coefficients (ICC). Overall ICC is 0.919 [0.892, 0.940] (excellent), confirming that violation rates reflect stable model behavior rather than API sampling noise. A notable pattern emerges: linguistic-layer types show uniformly excellent ICC

Table 4: Test–retest reliability (ICC) by ambiguity type.

Type	Layer	ICC	Interp.
scopal	ling.	0.989	excellent
lexical	ling.	0.981	excellent
coreferential	ling.	0.976	excellent
auth_scope	spec.	0.857	good
cond_prec	spec.	0.856	good
incompleteness	spec.	0.807	good
overall	—	**0.919**	**excellent**

Table 5: Judge calibration: LLM judge vs. human ground truth.

Metric	Value	Interpretation
Human inter-annotator κ	0.87	almost perfect
Judge–human F1	0.91	excellent
Precision	0.89	high
Recall	0.93	very high
Judge–human κ	0.84	almost perfect
GPT-5.4 judge F1	0.90	excellent
GPT-4.1 judge F1	0.92	excellent
Cross-judge ΔF1	0.02	negligible
Per-severity calibration		
High (clear violation)	F1=0.94	—
Mid (ambiguous judgment)	F1=0.88	—
Low (borderline)	F1=0.82	—

(0.976–0.989), while specification-layer types show good but lower ICC (0.807–0.857), consistent with the Disambiguation Channel Framework—specification-layer ambiguity forces agents to "guess" differently each run.

4.7 Judge Calibration

We validate the LLM-as-judge pipeline against human ground truth using 100 episodes annotated by two human experts (inter-annotator $\kappa = 0.87$, agreement 93%). Table 5 summarizes results. Judge–human F1 is 0.91, precision 0.89, recall 0.93, and $\kappa = 0.84$ (almost perfect). Calibration is strongest for clear violations (high-severity F1=0.94) and adequate for borderline cases (low-severity F1=0.82). Cross-judge asymmetry is negligible (ΔF1=0.02), confirming that the cross-model judging protocol does not introduce systematic bias.

For the failure-mode taxonomy (Section 6), 25 traces were coded by human experts using a refined codebook, yielding $\kappa = 0.76$ (good), upgrading from the automated inter-coder $\kappa = 0.594$ (moderate). This human anchor validates the qualitative findings.

5 Results

5.1 C1: Binary Effect — Ambiguity Universally Causes Safety Violations

Across all five models and six types, ambiguous policy specifications cause a +34.0pp increase in safety violation rates compared to matched unambiguous controls (pooled OR=15.1, 95% CI [10.78, 21.17], $p < 0.0001$; bootstrap CI [+31.2, +36.9]pp). Table 6 presents per-model results. The effect replicates without exception: the minimum per-model effect is +25.7pp (Claude), and the maximum is +39.0pp (DeepSeek), with all individual McNemar tests yielding $p < 10^{-17}$.

Table 6: C1: Binary effect of ambiguity on safety violations (5 models). OR = odds ratio.

Model	Δpp	OR	95% CI		p
Claude	25.7	8.92	5.61	14.19	$< 10^{-17}$
Qwen3	34.2	14.87	9.12	24.24	$< 10^{-18}$
GPT-4.1	34.7	15.31	9.56	24.52	$< 10^{-19}$
GPT-5.4	38.3	18.44	11.02	30.85	$< 10^{-20}$
DeepSeek	39.0	19.21	11.34	32.56	$< 10^{-17}$
Pooled	**34.0**	**15.1**	**10.78**	**21.17**	**<0.0001**

Table 7: Logistic regression predicting violation occurrence. Ambiguity condition dominates all other predictors.

Predictor	β	SE	z	p
condition_amb	2.278	0.116	19.65	<0.001
token_delta	0.022	0.006	3.49	0.0005
model_claude	−0.583	0.098	−5.95	<0.001
model_deepseek	0.412	0.095	4.34	<0.001
type_incomp	0.531	0.112	4.74	<0.001
type_coref	−0.447	0.118	−3.79	0.0002

Logistic regression confirms that the ambiguity condition is the dominant predictor ($\beta = 2.278$, $z = 19.65$, $p < 0.001$), dwarfing all other factors including token delta ($\beta = 0.022$, $z = 3.49$, $p = 0.0005$), model, and type (Table 7).

5.2 C2: Type Hierarchy — Three Tiers of Differential Nocuity

The omnibus χ^2 test confirms a significant type hierarchy ($\chi^2 = 28.02$, $p = 3.6 \times 10^{-5}$, Cramér's $V = 0.137$; power=0.993, MDE=0.093). Table 8 shows a three-tier structure confirmed by TOST analysis (8/15 pairs equivalent at ±15pp, 4/15 at ±10pp):

The 22pp gap between incompleteness (53.6%) and coreferential (31.6%) drives the effect. Although $V = 0.137$ is below the conventional "moderate" threshold (0.20), the study is well-powered (0.993) to detect effects as small as $V = 0.093$, confirming the hierarchy is real but modest—the three-tier structure captures the pattern better than any single summary statistic.

Coreferential ambiguity's low rank has a structural explanation: agents partially disambiguate via tool responses (a disambiguation channel), and conservative refusal rates are elevated (6–11% vs. 2–3% average), suggesting agents sometimes *pause* rather than guess.

5.3 C3: Layer Effect — Suggestive but Fragile

In aggregate, specification-layer ambiguities produce higher violation rates than linguistic-layer ambiguities (44.0% vs. 37.9%, $\Delta = 6.08$pp, $\chi^2 = 5.73$, $p = 0.017$, $V = 0.062$, OR=1.29 [1.05, 1.58]). Token delta partially mediates this effect: specification-layer clauses require more tokens to disambiguate (mean Δ_{tok}: 33.0 vs. 15.9).

However, within-judge subgroup analysis reveals fragility: the GPT-4.1-judged group (GPT-5.4, DeepSeek) shows $\Delta = +8.0$pp ($p = 0.059$, marginal), while the GPT-5.4-judged group (GPT-4.1, Claude, Qwen3) shows $\Delta = +4.8$pp ($p = 0.157$, n.s.). Per-model layer gaps vary dramatically: GPT-5.4 (+14.0pp), GPT-4.1 (+12.0pp), Claude (+3.3pp), DeepSeek (+2.0pp), and Qwen3 (−0.9pp, *reversed*). The aggregate layer effect is driven primarily by GPT models' elevated authorization-scope rates (46–64%), which inflate their specification-layer average.

We frame C3 as *suggestive*: specification-layer types are more dangerous on average, but the aggregate masks a model×type interaction (C4) that is the more fundamental finding.

Table 8: C2: Type hierarchy (ambiguous condition, pooled across 5 models).

Type	Layer	Viol. rate	Tier
incompleteness	spec.	53.6%	high
lexical	ling.	42.8%	mid
auth_scope	spec.	41.6%	mid
scopal	ling.	39.4%	mid
cond_prec	spec.	36.8%	mid
coreferential	ling.	31.6%	low

Table 9: C4: Model × type interaction (violation rates, ambiguous condition).

Type	GPT-5.4	GPT-4.1	Claude	Qwen3	DeepSeek
auth_scope	**64%**	46%	**12%**	26%	60%
cond_prec	34%	36%	36%	40%	38%
incompleteness	46%	58%	48%	62%	54%
lexical	42%	34%	32%	52%	54%
coreferential	18%	32%	22%	42%	44%
scopal	42%	38%	32%	37%	48%

5.4 C4: Model × Type Interaction — The Star Finding

The model main effect is highly significant ($\chi^2 = 31.02$, $p = 3.0 \times 10^{-6}$, $V = 0.102$). In the ambiguous condition alone, $\chi^2 = 24.01$, $p = 7.9 \times 10^{-5}$. Overall vulnerability ranks: Claude 30.3% < GPT-5.4 41.0% ≈ GPT-4.1 40.7% < Qwen3 43.1% < DeepSeek 49.7%.

Table 9 presents the full model×type interaction matrix. Three striking patterns emerge:

Authorization scope is the model-differentiating type. Its 52pp range (12% Claude to 64% GPT-5.4) is by far the largest. Claude's low vulnerability suggests its safety training specifically addresses authorization boundary reasoning.

Incompleteness is a safety-training blind spot. All models show high rates (46–62%, range only 16pp). When information is structurally absent, no amount of safety training compensates.

Conditional precedence is model-invariant. The 6pp range (34–40%) indicates all architectures handle explicit rule conflicts similarly.

Per-model χ^2 tests quantify type differentiation: GPT-5.4 ($V = 0.28$, $p = 0.0003$) and Claude ($V = 0.24$, $p = 0.003$) show the strongest type-specific profiles, Qwen3 moderate ($V = 0.23$, $p = 0.007$), GPT-4.1 borderline ($V = 0.18$, $p = 0.078$), and DeepSeek shows *no* significant type differentiation ($V = 0.14$, $p = 0.277$)—uniformly high vulnerability across all types.

Inter-model agreement is only fair (Fleiss $\kappa = 0.384$): models agree on the binary effect but disagree on *which specific clauses* trigger violations. Notably, 27% of ambiguous episodes show *systematic nocuity*—all models converge on the *same wrong* interpretation—suggesting a third category beyond Berry and Kamsties' nocuous/innocuous dichotomy (see Discussion).

6 Analysis: Failure Modes and Ambiguity Blindness

We analyze 300 violation traces (123 from GPT-5.4, 122 from GPT-4.1, 55 from other models) and identify four failure-mode categories. Table 10 shows the distribution for GPT-5.4.

Table 10: Failure mode distribution by ambiguity type (GPT-5.4). Values are proportions.

Type	Unauth. escal.	Assump. action	Scope misapp.	Arb. rule sel.
auth_scope	**.72**	.16	.08	.04
incompleteness	.18	**.61**	.14	.07
scopal	.04	.12	**.76**	.08
lexical	.08	**.71**	.15	.06
cond_prec	.12	.18	**.41**	.29
coreferential	.16	.28	.32	.24

Table 11: Ambiguity Blindness: meta-cognitive awareness in violation episodes.

Model	Awareness	Surface adopt.	Escalate to user	Conserv. refusal
GPT-5.4	4.1%	91.2%	0.8%	3.9%
GPT-4.1	10.7%	88.1%	0.4%	2.8%

Type-diagnostic signatures. Five of six types share the same dominant failure mode across GPT-5.4 and GPT-4.1: authorization scope → unauthorized escalation (72%/83%); incompleteness → assumption-based action (61%/66%); scopal → scope misapplication (76%/74%); lexical → assumption-based action (71%/53%). Only coreferential diverges, showing a flat distribution with no single dominant mode—reflecting the diversity of resolution strategies for referential ambiguity.

Layer mechanism. Linguistic-layer violations are dominated by scope misapplication (39% vs. 15% for specification-layer): the agent *misparses* the clause. Specification-layer violations are dominated by unauthorized escalation and assumption-based action: the agent *guesses and acts* when information is absent. This misparse-vs-guess dichotomy provides the qualitative *why* behind the quantitative layer gap.

Ambiguity Blindness. Agents acknowledge ambiguity in only 4.1% (GPT-5.4) and 10.7% (GPT-4.1) of violation episodes (Table 11). Surface adoption exceeds 88% across all types—agents accept the first plausible parse without flagging uncertainty. User escalation occurs in fewer than 1% of episodes. The frontier model (GPT-5.4) shows *lower* awareness than the mid-tier model (GPT-4.1), suggesting a possible inverse relationship between action-orientation and meta-cognitive caution.

Cross-domain consistency. Failure mode profiles are consistent across airline and retail domains ($\Delta < 13$pp), with the largest domain effect on unauthorized escalation (retail 38.6% vs. airline 26.1%), confirming that type-specific signatures generalize.

7 Discussion

Differential Nocuity. Berry and Kamsties' nocuous/innocuous dichotomy assumes ambiguity's harm depends on whether readers diverge. Our data extend this framework to LLM agents: all tested ambiguity types are nocuous (+34.0pp), but the *degree* of nocuity varies by type (3-tier hierarchy) and the *pattern* of vulnerability varies by model (52pp range on authorization scope). We propose the term *differential nocuity* for this phenomenon: the safety impact of ambiguity is jointly determined by type and model, not a property of the ambiguity alone.

Systematic Nocuity as a third category. In 27% of ambiguous episodes, all models converge on the same incorrect interpretation. This *systematic nocuity* is more dangerous than divergent nocuity—it is invisible in inter-model agreement metrics and appears "correct" until validated against ground truth. We propose this as an extension to Berry and Kamsties' framework.

Safety training selectivity. The model×type interaction reveals that safety training differentially addresses ambiguity types. Claude's dramatic auth_scope robustness (12%) against GPT-5.4's vulnerability (64%) suggests explicit or implicit training on authorization boundary reasoning. In contrast, incompleteness rates are uniformly high (46–62%)—when information is literally absent from the policy, safety training cannot compensate. DeepSeek's uniform vulnerability profile ($V = 0.14$, n.s.) contrasts with GPT-5.4 ($V = 0.28$) and Claude ($V = 0.24$), suggesting less type-specific safety training.

Disambiguation Channel Framework. The three-tier hierarchy and ICC pattern converge on an explanatory framework: linguistic ambiguity (coreferential, scopal) has partial *disambiguation channels*—tool responses, world knowledge, or structural cues that allow partial resolution. Specification ambiguity (authorization scope, incompleteness) has no external channel, forcing agents to guess. Evidence: (i) coreferential shows elevated conservative refusal (6–11%); (ii) linguistic-layer ICC is uniformly excellent (0.976–0.989) while specification-layer ICC is lower (0.807–0.857); (iii) the failure mode split (misparse vs. guess-and-act) mirrors this distinction.

Practical implications. Effective mitigation requires two complementary strategies: (1) *type-aware policy auditing*—incompleteness and authorization scope should receive priority attention, as they pose the greatest and most variable risk; (2) *model-specific risk assessment*—the same policy may be safe for one model but dangerous for another, particularly for authorization scope. The DeBERTa classifier proof-of-concept (binary F1=0.81; Appendix) suggests automated type-aware auditing is feasible, with per-type detection difficulty mirroring the disambiguation channel hierarchy (conditional precedence F1=0.94 vs. incompleteness F1=0.34).

8 Conclusion

We present the first causal, type-granular, cross-model study of policy ambiguity's impact on LLM agent safety. Three key findings emerge from 3,000 agentic episodes across five models. First, policy ambiguity universally causes massive safety violations (+34.0pp), with matched-pair design and perturbation controls providing the strongest causal evidence in the agent safety literature. Second, vulnerability is not uniform: a three-tier type hierarchy (incompleteness $\gg$ mid cluster $\gg$ coreferential) interacts with model architecture, producing dramatically different vulnerability profiles—authorization scope ranges 52pp across models while incompleteness remains universally dangerous. Third, agents exhibit pervasive Ambiguity Blindness, acknowledging ambiguity in fewer than 11% of violation episodes. These findings establish differential nocuity as a framework for understanding and mitigating ambiguity-driven safety risks, and argue that both type-aware auditing and model-specific risk assessment are necessary for safe agent deployment.

Limitations

Our study has three main limitations. First, evaluation covers two domains (airline and retail); generalization to healthcare, legal, or financial policies remains untested, though cross-domain failure-mode consistency ($\Delta < 13$pp) is encouraging. Second, clause pairs are synthetically constructed, and while linguistic property comparisons with DRIP-R natural ambiguities show no

significant differences (all KS $p > 0.10$), real-world policies may exhibit more complex ambiguity interactions. Third, while the aggregate layer effect (C3) reaches significance ($p = 0.017$), within-judge subgroup analyses are fragile ($p = 0.059$ and $p = 0.157$), indicating that the specification-vs-linguistic distinction may be model-dependent rather than fundamental.

A Ambiguity Detection Classifier

As a preliminary proof of concept for automated policy auditing, we fine-tune DeBERTa-v3-large on our 300 clause pairs (augmented to 500 clauses) using 5-fold cross-validation. Binary ambiguity detection achieves F1=0.81 (precision 0.79, recall 0.83, FPR 4.4%). Six-type classification achieves macro F1=0.63.

Per-type detection difficulty mirrors the Disambiguation Channel Framework: conditional precedence (F1=0.94) has explicit conflict markers; scopal (0.75) and coreferential (0.70) have structural patterns; authorization scope (0.58) is context-dependent; lexical (0.39) requires world knowledge; and incompleteness (0.34) requires detecting *absence*—what is not there.

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